

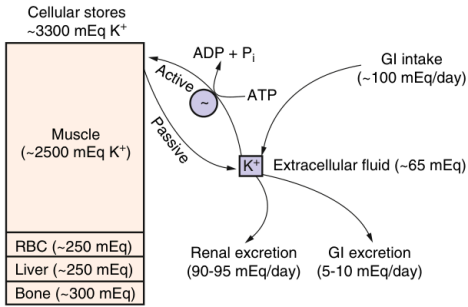


Fig. 16.1 Body K⁺ distribution and cellular K⁺ flux. (From Wingo CS, Weiner ID. Disorders of potassium balance. In: Brenner BM, ed. *The Kidney*. vol 1. WB Saunders; 2000:998–1035.)

Fig_16_1

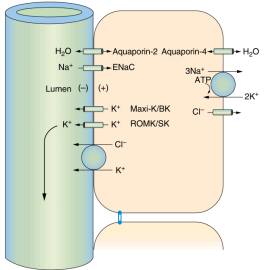


Fig. 16.4 K⁺ secretory pathways in principal cells of the connecting segment (CNT) and cortical collecting duct (CCD). The absorption of Na⁺ via the amiloride-sensitive epithelial sodium channel (ENaC) generates a lumen-negative potential difference, which drives K⁺ excretion through the apical secretory K⁺ channel ROMK. Flow-dependent K⁺ secretion is mediated by an apical voltage-gated, calcium-sensitive BK channel. Chloride-dependent, electroneutral K⁺ secretion is likely mediated by a K⁺-Cl⁻ cotransporter. Water transport in principal cells occurs via aquaporin-2 (Aqp-2) and aquaporins-3/4 (Aqp-3/4).

Fig_16_4

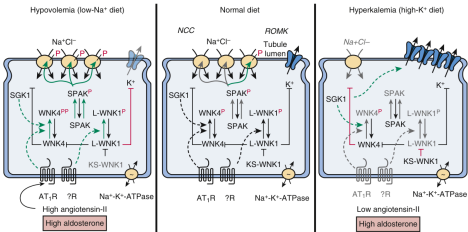


Fig. 16.7 Integrated regulation of Na⁺-Cl⁻ and K⁺ transport in the DCT, CNT, and CCD. Green arrowheads (activating pathways), red blunt end (inhibitory pathway). The left panel shows the pathway in the setting of a low-Na⁺ diet, wherein Ang II and SGK-1 signaling leads to phosphorylation of WNK4. This stimulates phosphorylation of SPAK, which in turn phosphorylates and activates thiazide-sensitive Na⁺-Cl⁻ cotransport in the DCT via NCC. Stimulation of unknown receptors is hypothesized to cause phosphorylation of L-WNK1, which can also stimulate SPAK phosphorylation. L-WNK1 has other functions: 1. It blocks the NCC-inhibitory form of WNK4, thus activating NCC; and 2. It inhibits secretion of K⁺ via ROMK channels. The right panel shows the pathway in the setting of high dietary K⁺ intake, wherein aldosterone is stimulated and Ang-II is low. In the absence of sufficient Ang-II, AT1R cannot activate WNK4. This reduces SPAK activation and NCC phosphorylation. Dietary potassium loading also increases the level of KS-WNK1 isoform to suppress the activity of L-WNK1. In consequence, the inhibitory effect of WNK4 on NCC dominates, blocking traffic of NCC to the apical membrane and thereby reducing NCC activity. KS-WNK1 also blocks the effect of L-WNK1 on ROMK endocytosis, causing ROMK to increase at the apical membrane. The net effect is that K⁺ secretion in the DCT and CNT/CCD is maximized, whereas NCC is suppressed. Aldosterone stimulation of ENaC, not shown, offsets the decreased Na⁺ reabsorption by NCC, allowing robust potassium secretion without changes in sodium balance. The roles of WNK3, SGK1, and c-src cytoplasmic tyrosine kinases are not shown in the interest of clarity; see text for further details. NCC, Na⁺-Cl⁻ cotransporter; ROMK, renal outer medullary K channel; SPAK, STE20/BPS1-related protein/kinase; WNK, WNK-Nipkin-related kinase. (From Weisinger PA, Chang YF, Delgado E, Wade JB, Multigene kinase network, kidney transport, and salt in essential hypertension. *Kidney Int*. 2010;77:1063–1069.)

Fig_16_7

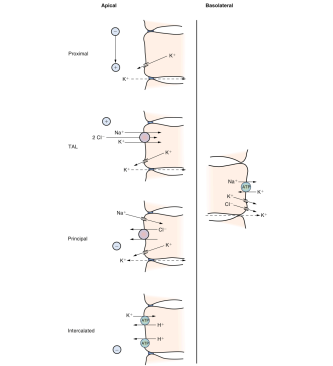


Fig. 16.2 Schematic cell models of potassium transport along the nephron. Cell types are as specified; note the differences in luminal potential difference along the nephron. TAL, Thick ascending limb; P, Proximal; D, Distal; I, Interstitial. (From Weisinger PA, Chang YF, Delgado E, Wade JB, Multigene kinase network, kidney transport, and salt in essential hypertension. *Kidney Int*. 2010;77:1063–1069.)

Fig_16_2

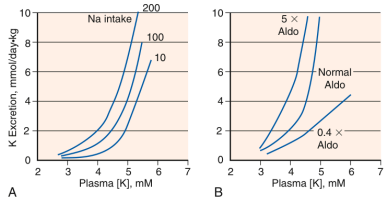


Fig. 16.5 (A) Relationship between steady-state plasma K⁺ and urinary K⁺ excretion in the dog, as a function of dietary Na⁺ intake (mmol/day). Animals were adrenalectomized and replaced with aldosterone, and dietary K⁺ and Na⁺ content were varied as specified. (B) Relationship between steady-state plasma K⁺ and urinary K⁺ excretion as a function of circulating aldosterone. Animals were adrenalectomized and variably replaced with aldosterone, dietary K⁺ content was varied. (A from Young DB, Jackson TE, Tipayamont U, Scott RC. Effects of sodium intake on steady-state potassium excretion. *Am J Physiol*. 1984;246:F772–778. B from Young DB. Quantitative analysis of aldosterone's role in potassium regulation. *Am J Physiol*. 1988;255:F811–822.)

Fig_16_5

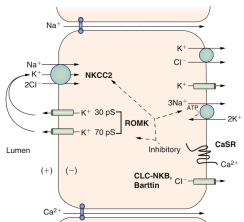


Fig. 16.3 Bartter syndrome and the thick ascending limb. Bartter syndrome can result from loss-of-function mutations in the Na⁺-K⁺-2Cl⁻ cotransporter NKCC2, the K⁺ channel subunit ROMK, or the Cl⁻ channel subunits CLC-NKB and Barttin (Bartter syndrome types I to IV, respectively). Gain-of-function mutations in the calcium-sensing receptor CaSR can also cause a Bartter syndrome phenotype (type V); the CaSR has an inhibitory effect on salt transport by the thick ascending limb, targeting several transport pathways. ROMK encodes the low conductance 30 pS K⁺ channel in the apical membrane and also appears to function as a critical subunit of the higher conductance 70 pS channel. The loss of K⁺ channel activity in Bartter syndrome type II leads to reduced apical K⁺ recycling and reduced Na⁺-K⁺-2Cl⁻ cotransport. Decreased apical K⁺ channels also lead to a decrease in the lumen-positive potential difference, which drives paracellular Na⁺, Ca²⁺, and Mg²⁺ transport.

Fig_16_3

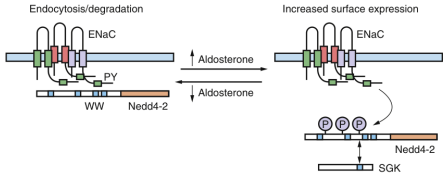


Fig. 16.6 Coordinated regulation of ENaC by the aldosterone-induced SGK kinase and the ubiquitin-ligase Nedd4-2. Nedd4-2 binds via its WW domains to ENaC subunits via their “PPXY” domains (denoted PY here), ubiquitinating the channel subunits and targeting them for removal from the cell membrane and destruction in the proteasome. Aldosterone induces the SGK kinase, which phosphorylates and inactivates Nedd4-2, thus increasing surface expression of ENaC channels. Mutations that cause Liddle syndrome affect the interaction between ENaC and Nedd4-2. (From Snyder PM, Olson DR, Thomas BC. Serum and glucocorticoid-regulated kinase modulate Nedd4-2-mediated inhibition of the epithelial Na⁺ channel. *J Biol Chem*. 2002;277:5–8.)

Fig_16_6

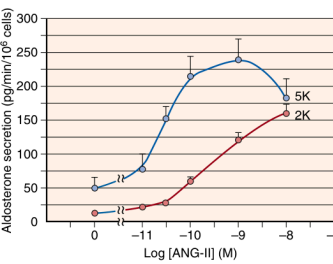


Fig. 16.8 Synergistic effect of increased extracellular K⁺ and angiotensin-II (ANG-II) in inducing aldosterone release from bovine adrenal glomerulosa cells. Dose-response curves for ANG-II were performed at extracellular K⁺ of 2 mmol/L (○) and 5 mmol/L (●). (From Chen XL, Bayliss DA, Fern RJ, Barrett PQ. A role for T-type Ca²⁺ channels in the synergistic control of aldosterone production by ANG II and K⁺. *Am J Physiol*. 1999;276:F674–683.)

Fig_16_8



Fig. 16.9 Hyperkalemic periodic paralysis (HPPV) due to mutations in the voltage-gated Na⁺ channel of skeletal muscle. (A) The disorder is particularly common in American Quarter Horses; an affected horse is shown during a paralytic attack, triggered by rest after heavy exercise. (B) Mechanistic explanation for muscle paralysis in HPPV. (A, Coxsey D, Ellis D, Hoffman SL, From Latham RW, Latham RW. Voltage-gated Na⁺ channels and hereditary disease. *Physiol Rev*. 1999;79:1317–1372.)

Fig_16_9

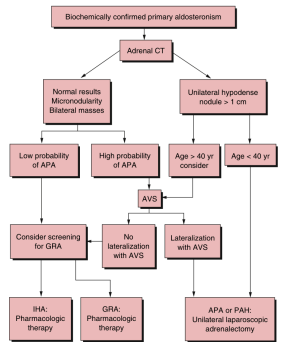


Fig. 16.10 Diagnostic algorithm in patients with primary hyperaldosteronism. Adrenal adenoma (APA) must be distinguished from glucocorticoid-remediable hyperaldosteronism (PHA) or GRA, primary or unilateral adrenal hyperplasia (PAH), and idiopathic hyperaldosteronism (IHA). This requires computed axial tomography (CT), adrenal venous sampling (AVS), and the relevant diagnostic biochemical and hormonal assays (see text). (From Young WF Jr. Adrenalectomy for primary aldosteronism. *Ann Intern Med*. 2003;139(2):157–159.)

Fig_16_10

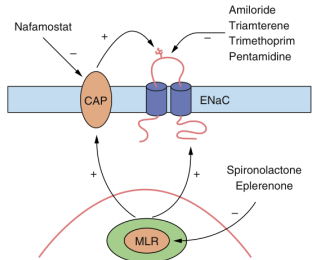


Fig. 16.13 Pharmacologic inhibition of the epithelial Na⁺ channel ENaC. Whereas amiloride and related compounds directly inhibit the channel, the protease inhibitor nafamostat inhibits hemibrane-associated proteases such as CAP1, thus indirectly inhibiting the channel. Spironolactone and related drugs inhibit the mineralocorticoid receptor, thus reducing transcription of the α -subunit of ENaC, the ENaC-activating kinase SGK, and several other target genes (see text for details).

Fig_16_13

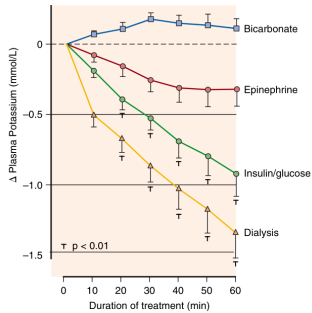


Fig. 16.16 Changes in serum K⁺ during intravenous infusion of bicarbonate, epinephrine, or insulin in glucose, and during hemodialysis. (From Blumberg A, Weidmann P, Shaw S, Gnadinger M. Effect of various therapeutic approaches on plasma potassium and major regulating factors in terminal renal failure. *Am J Med*. 1988;85:507–512.)

Fig_16_16

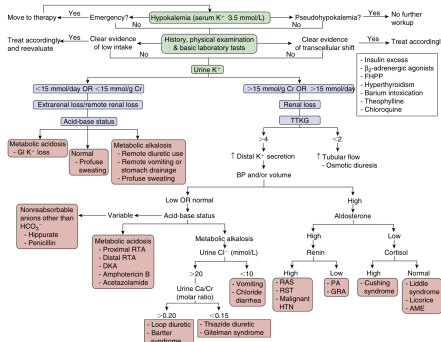


Fig. 16.11 The clinical approach to hypokalemia. See text for details. AME, Apparent mineralocorticoid excess; BP, blood pressure; CCG, cortical collecting duct; DKA, diabetic ketoacidosis; PHPP, familial hyperkalemic periodic paralysis; GI, gastrointestinal; GRA, glucocorticoid-remediable aldosteronism; HTN, hypertension; PA, primary aldosteronism; LALS, latent alveolar osteitis; RST, renal-secreting tumor; RTA, renal tubular acidosis; TTGG, transubular potassium gradient.

Fig_16_11

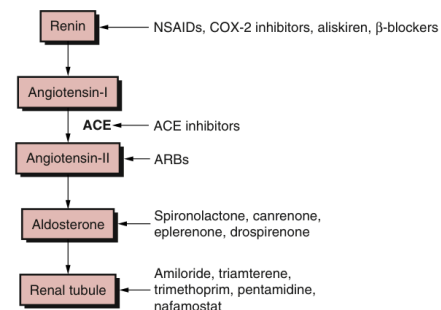


Fig. 16.14 Medications that target the renin-angiotensin-aldosterone axis are common causes of hyperkalemia, as are drugs that inhibit epithelial Na⁺ channels (ENaC) in the renal tubule (CNT or CCD).

Fig_16_14

Table 16.1 Factors Affecting K ⁺ Distribution Between Intracellular and Extracellular Compartments	
Acute	
Factor	Effect on Potassium
Insulin	Enhanced cell uptake
β-Catecholamines	Enhanced cell uptake
α-Catecholamines	Impaired cell uptake
Acidosis	Impaired cell uptake
Alkalosis	Enhanced cell uptake
External potassium balance	Loose correlation
Cell damage	Impaired cell uptake
Hyperosmolality	Enhanced cell efflux
Chronic	
Factor	Effect on ATP Pump Density
Thyroid	Enhanced
Adrenal steroids	Enhanced
Exercise (training)	Enhanced
Growth	Enhanced
Diabetes	Impaired
Potassium deficiency	Impaired
Chronic renal failure	Impaired

From Gleibisch G. Renal potassium transport: mechanisms and regulation. *Am J Physiol*. 1998;274:F817–833.

Table_16_1

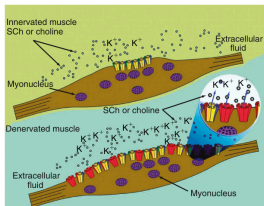


Fig. 16.12 Succinylcholine-induced efflux of potassium is increased in denervated muscle. In innervated muscle, succinylcholine interacts with the entire plasma membrane but depolarizes only the junctional (α1, β1, γ, and ε-multicoupled acetylcholine receptors (AChRs); this leads to a modest, transient hyperkalemia. With denervation, there is a considerable upregulation of muscle AChRs, with increased extrajunctional AChRs (α1, β1, γ, and ε-multicoupled) and acquisition of homomeric, neuronal-type α7-AChRs. Depolarization of denervated muscle leads to an exaggerated K⁺ efflux, due to the upregulation and redistribution of these AChRs. In addition, choline generated from metabolism of succinylcholine maintains the depolarization mediated via α7-AChRs, thus enhancing and prolonging the K⁺ efflux after paralysis has subsided. (From Marilyn JA, Richfield M. Succinylcholine-induced hyperkalemia in acquired pathologic states: etiologic factors and molecular mechanisms. *Anesthesiology*. 2006;104:158–169.)

Fig_16_12

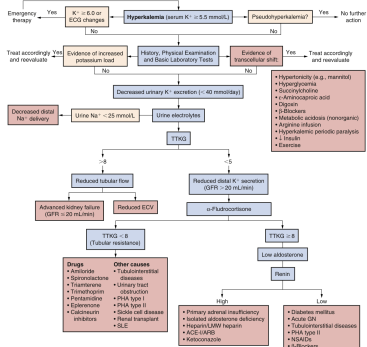


Fig. 16.15 Clinical approach to hyperkalemia. See text for details. ACE, Angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; CCG, cortical collecting duct; ECG, electrocardiogram; ECV, effective circulating volume; GRA, glucocorticoid-remediable aldosteronism; HTN, hypertension; LALS, latent alveolar osteitis; NSAIDs, nonsteroidal anti-inflammatory drugs; PHPP, pseudohyperaldosteronism; SLE, systemic lupus erythematosus; TTGG, transubular potassium gradient.

Fig_16_15

Table 16.2 Approximate Relationship Between Hyperkalemic Electrocardiographic Changes and Serum K ⁺	
Serum K ⁺	Electrocardiogram Abnormality
Mild hyperkalemia 5.5–6.5 mmol/L	Tall peaked T-waves with narrow base, best seen in precordial leads
Moderate hyperkalemia 6.5–8.0 mmol/L	Peaked T-waves Prolonged PR interval Decreased amplitude of P waves
Moderate hyperkalemia >8.0 mmol/L	Widening of QRS complex Absence of P wave Intraventricular blocks, fascicular blocks, bundle branch blocks, QRS axis shift Progressive widening of the QRS complex “Sine-wave” pattern (sinusoidal rhythm), ventricular fibrillation, asystole

From Mattu A, Brady WJ, Robinson DA. Electrocardiographic manifestations of hyperkalemia. *Am J Emerg Med*. 2000;10:721–729.

Table_16_2

Table 16.3 Foods With High Potassium Content
Highest content (>1000 mg [25 mmol]/100 g) Dried figs Molasses Seaweed
Very high content (>500 mg [12.5 mmol]/100 g) Dried fruits (dates, prunes) Nuts Avocados Bran cereals Wheat germ
Lima beans High content (>250 mg [6.2 mmol]/100 g) Vegetables Spinach Tomatoes Broccoli Winter squash Beets Carrots Cauliflower Potatoes
Fruits Bananas Cantaloupe Kiwi Oranges Mangos
Meats Ground beef Steak Pork Veal Lamb
From Gennari FJ. Hypokalemia. <i>N Engl J Med</i> . 1996;339:451–455.

Table_16_3